Names _____ School _____

_____ Team # _____ Score: _____ / 110

Chemistry Lab Test

created by kenniky

Part I. Thermodynamics [60 pt]

1. What are the four laws of thermodynamics? (6 pt)

   (0)

   (1)

   (2)

   (3)

2. A spherical balloon 30.0 cm in diameter is expanded until its diameter is 40.0 cm. If the pressure on the balloon is 1.00 atm, how much work was done on the balloon? (5 pt)

3. What is ΔH for the system in question 2 if ΔE is +4.00 J? (1 atm cm^3 = 0.1 J) (3 pt)

4. From questions 2 and 3, is the system endothermic or exothermic? (1 pt)

5. Find ΔH^o for the reaction

   $$2CO_2\,(g) + H_2O\,(g) \rightarrow C_2H_2\,(g) + 5/2\ O_2\,(g),$$

   given the following reactions and subsequent ΔH^o values:

   $C_2H_2\,(g) + 2H_2\,(g) \rightarrow C_2H_6\,(g)$ $\quad\Delta H = -94.5$ kJ
   $H_2O\,(g) \rightarrow H_2\,(g) + ½\ O_2\,(g)$ $\quad\Delta H = +71.2$ kJ
   $C_2H_6\,(g) + 7/2\ O_2\,(g) \rightarrow 2CO_2\,(g) + 3H_2O\,(g)$ $\quad\Delta H = -283$ kJ (4 pt)

6. For a particular reaction, $\Delta H = +145$ kJ. If the temperature is 273 K, for what values of ΔS is the reaction spontaneous? (3 pt)

7. A 10.00 mL container of water is at 10.0 ºC. A 2.23 g piece of zinc at 121 ºC is dropped into the container. What is the final temperature of the system? ($c_{Zn} = 0.384$ J/g ºC) (5 pt)

8. After the temperature of the system in question 7 has settled, the zinc is removed and heated to 98.6 ºC along with a 0.441 g piece of titanium. Both metals are then released back into the water simultaneously. What is the final temperature of this new system? ($c_{Ti} = 0.523$ J/g ºC) (8 pt)

9. At 273 K, the reaction

$$A\ (aq) \Leftrightarrow 2B\ (aq)$$

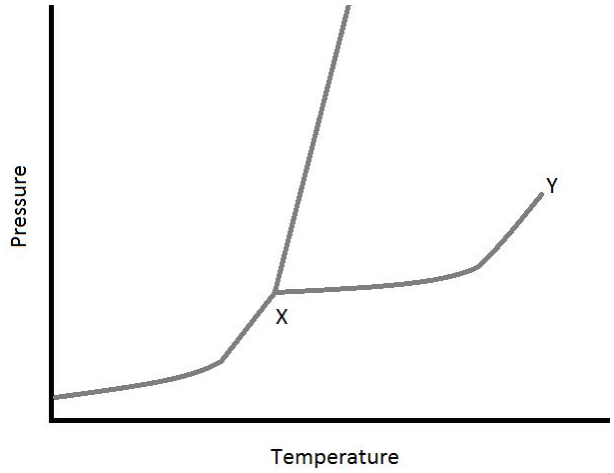
has an equilibrium constant of 375. What is ΔG^o for the reaction? (4 pt)

10. At 298 K, the equilibrium constant of the reaction in question 9 is 525. What are ΔH^o and ΔS^o for the reaction? (6 pt)

11. The reaction in question 9 is in equilibrium when half the amount of existing B is removed from the system. If the temperature is 400.0 K, what is ΔG? (7 pt)

12. You have a 1.00 mol sample of water at -30.0 ºC and you heat it until you have gaseous water at 140.0 ºC. Calculate q for the entire process. Use the following data:

c_{ice}	2.03 J/g ºC	
c_{water}	4.18 J/g ºC	
c_{steam}	2.02 J/g ºC	
H_2O (s) → H_2O (l)	ΔH_{fusion} = 6.02 kJ/mol	
H_2O (l) → H_2O (g)	$\Delta H_{vaporization}$ = 40.7 kJ/mol	(5 pt)



13. Above is my expertly drawn phase diagram. Label the three regions with the appropriate phase (solid, liquid, or gas), and below name and define the points X and Y. (5 pt)

X:

Y:

Part II. Physical Properties [50 pt]

1. At what temperature is water the most dense? (2 pt)

2. Which two elements are liquid at room temperature? (2 pt)

3. Graphite and diamond are both made of pure carbon; however, graphite is very soft while diamond is one of the hardest substances known. Explain. (4 pt)

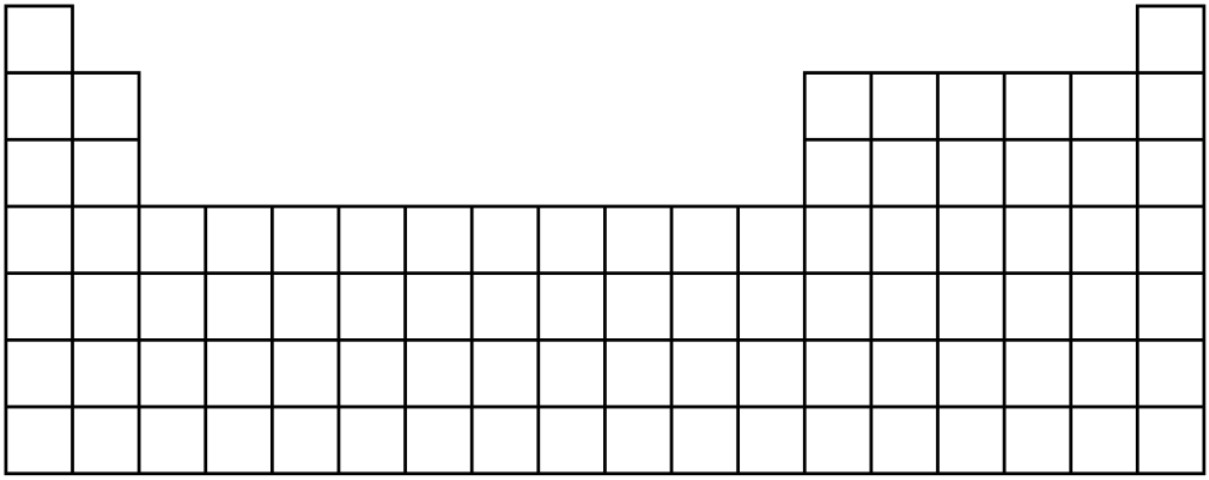
4. Explain the difference between an intensive and an extensive property, and give an example of each. (3 pt)

5. Why is salt soluble in water, but sugar not? (4 pt)

6. You are given samples of two unknown liquids, A and B. Create a quantifiable experiment to determine their relative viscosities. (4 pt)

7. In the event Bungee Drop, a weight, attached to an elastic cord, is dropped from a certain height. Say a 10 m cord with a 3 cm diameter is used with a 400 g weight. After the weight is released, it hangs 10.01 m below the release point. If a 600 g weight was used instead, how low would it hang? (7 pt)

8. On the blank periodic table below, use the letters below to mark where the elements are that have: (8 pt)
    A. the largest atomic radius
    B. the highest ionization energy
    C. the highest electron affinity
    D. the largest molar mass



9. Explain why each of the trends that you described above exists. (16 pt)
    a. Atomic radius

    b. Ionization energy

    c. Electron affinity

    d. Molar mass

SSSS - Repaired practice edition

Division C | Chemistry Lab | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

II-5. (6 points) Authored replacement for source item Question II-5: Solve this case and explain your reasoning: React 3.0 mol H_2 with 1.0 mol O_2. Find water and leftover hydrogen.

Answer: _____
_____
_____